Marçal Herraiz Bayó

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Website · LinkedIn · GitHub

Education

Present
Sept. 2025

PSL University — Paris, France
MSc in Mathematics — GPA: 15.052/20
Coursework at Dauphine and École Normale Supérieure.
- Ranked 5th of 116 students, top 5%.
- Full scholarship awarded by the Fondation Sciences Mathématiques de Paris.

June 2025
Sept. 2020

University of Barcelona — Barcelona, Spain
BSc in Mathematics — GPA: 9.02/10
- Ranked 5th of 141 students, top 4%.
- Thesis on flat connections and principal bundles. — Grade: 10/10

June 2025
Sept. 2020

University of Barcelona — Barcelona, Spain
BSc in Physics — GPA: 8.79/10
- Ranked 16th of 196 students, top 8%.
- Thesis on classical shadows and quantum computing. — Grade: 9.7/10

Work Experience

Present
July 2025

Westinghouse Nuclear — Madrid, Spain
Simulation Modeling Engineer
- Developing a new liquid-vapor-air multiphase model with compressibility terms, revised state variables, and a reformulated pressure solver.
- Presented preliminary results to department leadership, leading to approval of a new multiphase-modeling project.
- Built and validated a condenser heat-transfer model accounting for liquid level, geometry, tube coverage, and subcooling, with under 2% error against plant data.
- Resolved major energy-conservation errors in open tanks by implementing a new overflow model.
- Delivered validated thermal-hydraulic models to production software used in ongoing projects and plants under construction.

Research Experience

June 2025
Nov. 2024

Barcelona Supercomputing Center – BSC — Barcelona, Spain
Visiting Student
- Implemented a Shadow–Grouping algorithm combining classical shadows with observable grouping to improve quantum measurement efficiency.
- Benchmarked the method on H_2 and LiH Hamiltonians using PennyLane.
- Improved energy-estimation accuracy by up to 18-fold with two to three orders of magnitude fewer measurements than standard classical shadows.
- Supervisors: Prof. Dr. Bruno Julià, Dr. Berta Casas, Dr. Sergi Masot.

Institute for Research in Biomedicine – IRB Barcelona, Spain

Feb. 2025 – July 2024

Research Intern

- Modeled DNA flexibility using coarse-grained computational methods in C.
- Extended the model to analyze how DNA damage affects flexibility.
- Developed Python algorithms to compute DNA geometric descriptors.
- Supervisor: Prof. Dr. Modesto Orozco.

Center for Mathematical Research – CRM Barcelona, Spain

Aug. 2023 – June 2023

Research Intern

- Studied bifurcation phenomena in the complex exponential family, focusing on the non-structural stability of $\exp(z)$.
- Supervisors: Prof. Dr. Núria Fagella, Prof. Dr. Xavier Jarque.

Honors and Awards

- **2025** FSMP Scholarship for postgraduate studies at PSL, among more than 1,000 candidates.
- **2024** Collaboration Scholarship for research toward the Mathematics BSc thesis.
- **2024** Maths4Life Award, one of five IRB grants in mathematics and biomedicine.
- **2024** CRM Award for a Bachelor's thesis supervised by a CRM-affiliated researcher; declined.
- **2021–2025** Excellent with Honors in 16 subjects across the two BSc programs at UB.
- **2020** Honors in Baccalaureate for outstanding academic performance in high school.
- **2019** Mad for Science Programme, selective Physics course introducing students to research.

Selected Presentations

- **2025** Research Seminar at BSC: presented thesis work on Shadow–Grouping for measurement-efficient quantum energy estimation.
- **2024** Extraordinary BioMed Seminar – Maths4Life: presented research on coarse-grained modeling of DNA flexibility.
- **2023** Barcelona Introduction to Mathematical Research: presented work on bifurcation phenomena in the complex exponential family.

Languages

Spanish	Native
Catalan	Native
English	C1
French	A2

Coding Skills

Proficient	Python, C, Fortran, LaTeX.
Familiar	R, Bash, Mathematica, SageMath, HTML, CSS.
Libraries	NumPy, SciPy, Pandas, Scikit-learn, Matplotlib, PennyLane.
Tools	Git/GitHub, Unix/Linux, Jupyter.